



GPUS and NVIDIA Stock, 2019-2022

NVIDIA and the GPU Market

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What is NVIDIA?

- ▶ American technology company
- ▶ Founded 1993
- ▶ GPUs instead of CPUs
- ▶ First major success in 1999 selling GPUs to Xbox console
- ▶ Continued focusing on GPUs for gaming
- ▶ Began diversification through markets such as AI and Parallel Computing
- ▶ Continued making new models, GPUs remains primary focus, but for multiple industries
- ▶ Market Dominance on GPUs



What is a GPU?

- ▶ Graphics Processing Unit
- ▶ Alternative to the CPU
- ▶ Originally designed for graphics rendering and video game performance
- ▶ Have many less powerful cores instead of a few very powerful cores
- ▶ Optimized for many small tasks that can be run in parallel
- ▶ Speed noticed by researchers
- ▶ Used today in robotics, crypto, video editing, automotives, and Machine Learning
- ▶ Can be expensive to manufacture



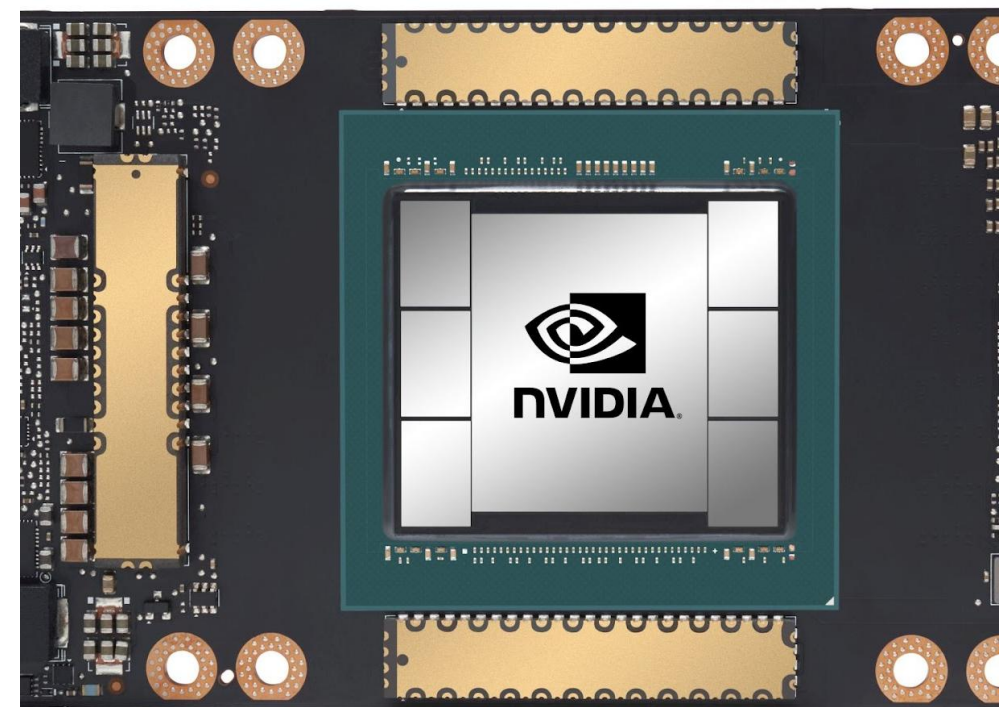
2019 - Acquisitions and Recovery

- ▶ 2018 was a bad year for Nvidia with share price halving in Q4
- ▶ Purchased Mellanox Technologies, high-performance computing
- ▶ Temporary decline in shares due to crypto hype dying down
- ▶ GeForce 16 Series and RTX 2070 deployments massive success
- ▶ Shares rise from \$35 to \$60
- ▶ Overall a “normal” and good rebound year for Nvidia



2020 - COVID: A Blessing and a Curse

- ▶ COVID-19 Pandemic creates new challenges and opportunities for Nvidia
- ▶ Lockdowns increase demand for virtual products
- ▶ Deployment of new Ampere graphics cards in the RTX 3000 Series
- ▶ Chip shortage (GeForce30) with supply chain issues
- ▶ Crypto scalpers buy up GPUs as soon as they are sold
- ▶ In August announcement to purchase ARM, their main competitor in GPU production
- ▶ Metaverse hype leads to boost in sales
- ▶ Good year for Nvidia but not GPU consumers



Still Waiting

The gap between ordering a chip and getting it is growing



Source: Susquehanna Financial Group

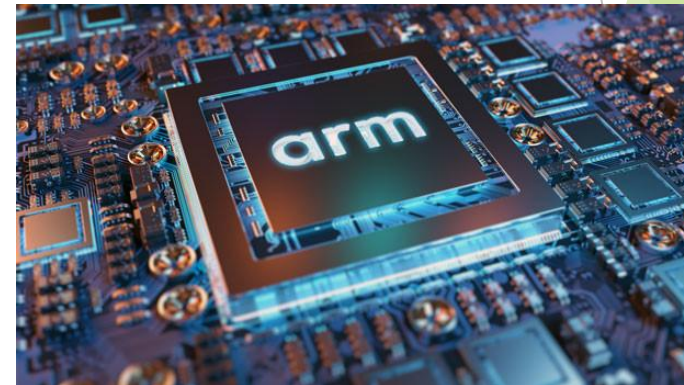
2021 - High Demand Yields High Growth

- ▶ Domination of GPU market continues, rising to over 80% market share
- ▶ Ampere card performance expanded as new models released such as RTX 3080
- ▶ Expand into new markets with Canvas tool, Adobe partnership, GeForce Now, and more
- ▶ ARM acquisition under scrutiny from regulators
- ▶ US-China trade war strains GPU and semiconductor production even harder
- ▶ Crypto scalping and usage of GPUs increases
- ▶ Extremely high demand and lack of supply skyrockets stock by the end of 2021



2022 - Major Failures, Signs of Hope

- ▶ A Year of Controversies and Failures
- ▶ In Q1 deal to acquire ARM falls through due to “significant regulatory challenges”
- ▶ Ampere replaced by Lovelace GPUs were faster but had major problems
- ▶ Power adapter issues including melting
- ▶ RTX 4080 pricing expensive for not much change
- ▶ Despite negativity still holds almost 90% market share
- ▶ AI hype begins with the release of OpenAI’s ChatGPT
- ▶ Could this technology do something in the future?
- ▶ Shares overall still about half in value



References

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